DINUKA EKANAYAKE

Software Engineer

Full-stack software engineer with hands-on experience building scalable web applications, AR-based interfaces, and applied AI systems. Delivered production-grade features in ERP systems during industry internships and built an AI-powered virtual try-on platform integrating AR, segmentation, and diffusion models. Passionate about crafting user-focused digital experiences, solving complex engineering problems, and continuously expanding expertise across full-stack development and machine learning.

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WORK EXPERIENCE

Software Engineer

Csquare Technologies Private Limited | Jun 2023 - Jun 2025

Worked as a Software Engineer at Csquare Technologies Private Limited since June 2023. As an Intern - Software Engineer and Software Engineer, I had responsibilities to develop, maintain, and test ongoing web-based ERP systems developed by the company.

- **ERP Developments**: Designing, coding, and testing robust features and modules for companies' internal and external ERP systems.
- **System Analysis and Design**: Excelling in requirements gathering, translating complex business needs into clear, technical specifications, and designing scalable system architectures.
- **System Implementation**: Overseeing the successful deployment of new ERP solutions and updates, ensuring smooth transition and minimal disruption to operations.
- **Customer Support**: Providing expert, high-level technical support, resolving critical post-implementation issues, and ensuring high customer satisfaction.

Mainly working with : PHP, HTML, CSS, JS

SKILLS

Programming

Python, TypeScript, JavaScript, PHP, Java, C/ C++

Frontend Development

HTML5, Three.js, CSS, Tailwind CSS, Next.js, React

Backend Development & APIs

Node.js, Laravel, Flask, FastAPI, REST API Design & Integration

Databases

MySQL, MongoDB, Firebase

Cloud, DevOps & Deployment

AWS, Microsoft Azure, Docker, Git, GitHub, Vercel, Railway, Cloudinary, Hugging Face Spaces, CI/CD

AI / Machine Learning

TensorFlow/Keras, U^2-Net, MediaPipe Pose, OpenCV, Image Classification, Segmentation, Model Deployment

Tools & Development Workflow

VS Code, Figma, Adobe Photoshop, Android Studio, Agile/Scrum, Version Control

EDUCATION

BSc (Hons) in Engineering in Information and Communication Engineering (Software Engineering) Sri Lanka Technology Campus | 2019 - 2025

- Specializing Software Dev & Design | OOP | DS & Algorithms | DBMS | Cloud | Data Science | Telecommunication | ML DL NN

Advanced Level in Physical Science Stream Ranabima Royal College | 2008 - 2016

PROJECTS

Hybrid AR + AI Virtual Try-On System for Fashion E-Commerce

Built a dual-workflow virtual try-on platform that allows users to preview garments in real-time AR or generate high-fidelity AI try-on results from photos.
Designed a complete pipeline including garment classification (CNN), background removal (U^2-Net), AR overlay using MediaPipe Pose, and a diffusion-based virtual try-on engine (CatVTON) deployed on Hugging Face.

Tech: Next.js, FastAPI, Python, TensorFlow/Keras, U^2-Net, MediaPipe, Cloudinary, Gradio, Hugging Face Spaces

Interactive 3D Virtual Art Gallery (Web-Based)

I created a fully interactive 3D virtual art gallery where users can walk through a gallery space, view paintings in a realistic environment, and explore digitally placed sculptures. The system features custom room layouts, dynamic artwork placement, real-time navigation, textured walls/floors, and 3D statue models rendered using Three.js.

Tech: JavaScript, Three.js, WebGL, HTML/CSS

Moodify – Emotion-Based Music Recommendation Web App

I developed a web application that detects the user's emotional state and recommends personalized music playlists using real-time emotion analysis and the Spotify API. The app integrates a machine learning model with a Flask backend and features a responsive frontend for smooth user interaction.

Tech: Python, Flask, TensorFlow/Keras, JavaScript, HTML/CSS, Spotify API

Agri Mart Android Application Project

Agri Mart is an online shopping app that creates a digital market for all kinds of agricultural products. Everyone can sell their products and buy products through this application.

Tech: Java, Firebase

Database Management

I have created several databases related to my curriculum, which are included in the courses that I follow. I'm skilled at creating ER diagrams, normalizing data, and designing databases.

Tech: SQL, Nodejs, MongoDB

Web Development

I'm a self-learner in web development, and I have created several websites for learning. I have used HTML, CSS, JavaScript, PHP, and React to develop websites.

Tech: HTML, CSS, JavaScript, PHP, React

STRENGTHS

- Creative Problem-Solving
- Critical Thinking
- Leadership
- Flexibility
- Communication
- Team Collaboration
- Time Management
- Adaptability

VOLUNTEER EXPERIENCE

Vice Chairman
IEEE Computer Society Student Branch Chapter of SLTC | 2021 - 2022

Best Student Branch Chapter Recognition at IEEE Boost 2021 by IEEE Sri Lanka Section

Student Advisor - HackTron v1 .0 Project consisting
Phase 1 - 1O Day Game Development Bootcamp
Phase 2 - Inter-Uni Game Dev Hackathon (First Ever Hackathon by SLTC)
Phase 3 - RISE UP Industrial Talks

Student Advisor - Month of CodegeNiX 2021 Project which was a program series consisting of
1) Panel Discussion on How to become successful in IT Industry
2) Road to CODEMANIA 21' - Python Fundamentals and Algorithmic Workshop Series
3) AdaGrad 21' - Introduction to Machine Learning Workshop
4) CODEMANIA v2.0 - SLTC's First-Ever Algorithmic Programming Competition

Treasurer
Media Unit of SLTC | 2021 - 2022

Have completed several fundraising projects to collect funds for the Media Unit and held the position of Treasurer on several inter-university projects. Currently holding the position of Advisor to the Executive Committee.

Vice Chairman
CYBERHAT 1.0 project | Dec 2020 - jan 2021
IEEE Computer Society Student Branch Chapter of SLTC

"CYBERHAT 1.0 - Getting Started with Cybersecurity and Ethical Hacking" Project. The online workshop for those who are interested in cybersecurity & the ones who are searching for a cybersecurity career path.

Publicity Team Head
IEEE Computer Society Student Branch Chapter of SLTC | 2020 - 2021

Have completed several fundraising projects to collect funds for the Media Unit and held the position of Treasurer on several inter-university projects. Currently holding the position of Advisor to the Executive Committee.

Media and Publicity Team Deputy Head
Leo IEEE Sinhalen Docs Project | 2020 - 2021

An all-island youth empowerment project collaboratively done by the IEEE Student Branch of SLTC and the Leo Club of SLTC.

Promotional and Social Media Team Deputy Head
Green In ovation Sri Lanka - Project GISLA | 2020
IEEE Student Branch of SLTC

An all-island video documentary and digital art competition organized by the IEEE Student Branch of SLTC.

Photographer and Graphic Designer
Media Unit of SLTC | 2019 - 2022

Have covered more than 120 events as a photographer and worked as a graphic designer in media unit of SLTC.

REFERENCES

Mr. Chamara Rathnayake
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